MedVQA-Router: Bridging Binary Classification and Open-Ended Captioning for Medical Visual Question Answering

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Abstract

Medical Visual Question Answering (Med-VQA) requires simultaneous understanding of clinical images and natural-language questions. We address both the binary (yes/no) and open-ended subsets of Med-VQA using the `robailleo/medical-vision-llm-dataset` [12], which contains 4,793 samples across X-ray, CT, MRI, and ultrasound modalities. Our two-stage pipeline routes questions via a rule-based classifier, directing binary questions to a multimodal fusion model (ResNet-18 + BioBERT) and open-ended questions to GPT-5 Mini via the OpenAI API. For binary VQA we compare text-only baselines (TF-IDF + LR/SVM), an image-only CNN (ResNet-18), and fusion models. Our best binary classifier—ResNet-18 fused with fine-tuned BioBERT—achieves **79.5% accuracy** and **0.83 AUC-ROC**, outperforming single-modality baselines by over 7 percentage points. For the open-ended branch, we evaluate four vision-language models and find that GPT-5 Mini achieves the highest BERTScore F1 (0.7528) and Recall (0.8021), while InstructBLIP leads on ROUGE, demonstrating that GPT-5 Mini captures clinical meaning more completely than smaller fine-tuned models.

1 Introduction

Medical Visual Question Answering (VQA) has attracted significant research interest as a tool for automated clinical decision support. A Med-VQA system must jointly reason over a clinical image and a natural-language question to produce a correct answer—either binary ("Is there a fracture?") or open-ended ("What abnormality is visible?"), requiring different modelling strategies.

Medical datasets are limited in size due to the high cost of expert annotation and privacy regulations. Clinical images exhibit significant visual diversity across modalities, while questions use a highly specialized vocabulary. Standard vision-language models pretrained on natural images often fail to capture subtle pathological markers critical for accurate diagnosis.

We propose a two-stage pipeline that prioritizes task specialization: a routing classifier separates binary from open-ended questions, directing each to a dedicated model. For binary VQA, a multimodal fusion of ResNet-18 and BioBERT achieves 79.5% accuracy and 0.83 AUC-ROC, outperforming general-purpose models like CLIP. For open-ended captioning, we evaluate a progression of four vision-language models and find that GPT-5 Mini captures clinical meaning most completely, achieving the highest BERTScore F1 (0.7528) and Recall (0.8021).

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2 Background

Visual and Medical VQA. Since the VQA v2 dataset [3], VQA research has evolved from monolithic architectures to complex attention-based mechanisms. In the medical domain, datasets such as VQA-Med [5] and SLAKE [9] benchmark performance on radiology and pathology images, where accuracy is safety-critical and linguistic bias is a known challenge.

Transfer Learning and Biomedical Language Models. Due to the scarcity of annotated medical images, transfer learning from ImageNet-pretrained ResNet architectures [4] is standard practice. BERT [2] has spawned specialized variants such as BioBERT [6], pretrained on PubMed abstracts and PubMed Central full texts, making it significantly better suited to clinical terminology than general-domain models.

Vision-Language Models. CLIP [11] aligns images and text in a shared latent space via dual encoders, while BLIP [8] enables flexible generation through a multimodal encoder-decoder mixture. GPT-5 Mini extends this to large-scale instruction-following captioning via the OpenAI API.

3 Data

3.1 Dataset

We use the `robailleo/medical-vision-llm-dataset` [12] from HuggingFace, a combined medical vision-language dataset comprising **4,793 samples** drawn from two sources: ROCO (3,000 samples) and VQA-RAD (1,793 samples). The official split yields **3,834 training** and **959 validation** samples.

Modality distribution: X-ray dominates with 2,325 samples (48.5%), followed by CT (1,351; 28.2%), unknown modality (812; 16.9%), MRI (231; 4.8%), ultrasound (70; 1.5%), and microscopy/endoscopy (4; $<0.1\%$). **Body-part distribution:** Unknown (2,856), chest (718), abdomen (468), head (408), extremities (125), spine (119), and pelvis (99).

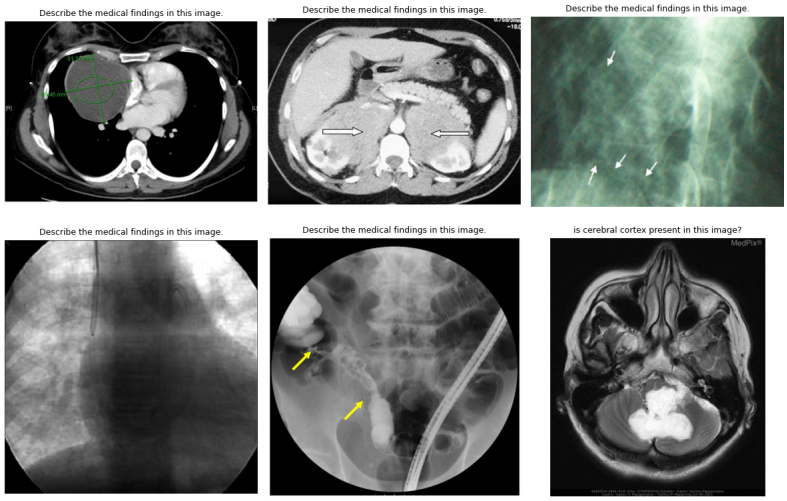

Figure 1: Example medical images from the dataset. The dataset includes multiple medical imaging modalities such as X-ray, CT, MRI, and ultrasound.

3.2 Preprocessing and Split

The preprocessing pipeline was designed to make the image and text data ready for the model. Each image is resized to 224×224 pixels so that all images have the same input size. Since some images are grayscale, every image is converted to three-channel RGB format. After that, the image is converted into a PyTorch tensor and normalized using ImageNet statistics with mean $[0.485, 0.456, 0.406]$ and standard deviation $[0.229, 0.224, 0.225]$. This is useful because the image encoder is based on pretrained CNN/ResNet models, which expect images in this format.

For the text part, the question is kept as the input sentence for the text encoder. Questions are tokenized with a maximum sequence length of 128 tokens. Longer questions are truncated, and shorter questions are padded when needed. This gives the model a fixed-size text input while still keeping the important information from the question.

The dataset already provides official train and validation splits, so we did not create a new random split for the main experiments. After filtering the dataset to only binary yes/no examples, the binary training set contained 750 samples and the binary validation set contained 190 samples. The class distribution stayed balanced in both splits. In the training split, there were 379 no examples and 371 yes examples. In the validation split, there were 94 no examples and 96 yes examples.

Because the predefined split was already balanced, we used it directly for training and evaluation. We also checked the label percentages in each split: the training split was about 50.53% no and 49.47% yes, while the validation split was about 49.47% no and 50.53% yes. This showed that the official split was reasonable for binary classification.

Table 1: Binary label distribution after filtering to yes/no answers. Label 0 represents "no" and label 1 represents "yes".

Split	Label 0 (No)	Label 1 (Yes)
Train	379	371
Validation	94	96

4 Methods

Our system has three components: (1) a routing classifier, (2) a binary VQA pipeline, and (3) an open-ended captioning pipeline.

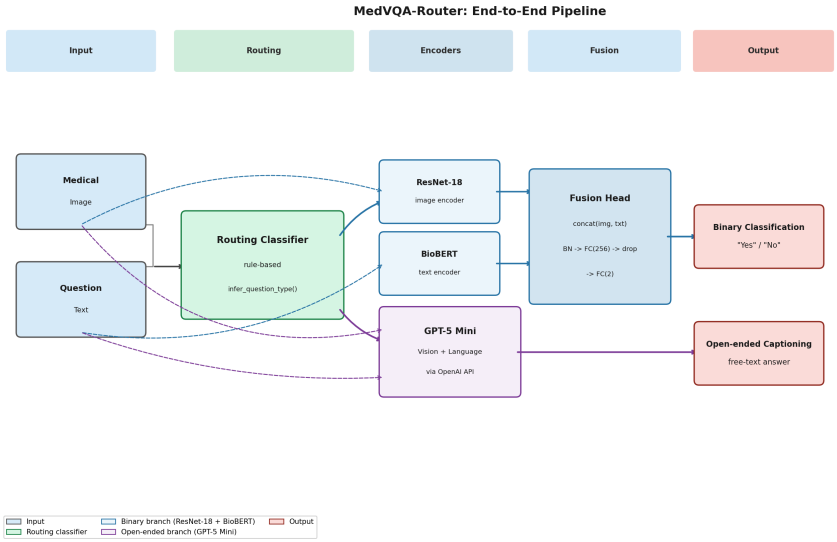

Figure 2: End-to-end pipeline: the routing classifier directs inputs to the binary classification or open-ended captioning branch.

4.1 Routing Classifier

A rule-based router in `src/utils.py` inspects each answer: "yes" or "no" answers are directed to binary classification; all others to open-ended captioning. The routing is deterministic and error-free by construction, operating on ground-truth answer labels across all 4,793 samples.

4.2 Text-Only Baselines

Questions are vectorized with TF-IDF (10,000 terms, `ngram_range=(1,3)`, `sublinear_tf=True`). We train Logistic Regression (L2, `max_iter=1000`) and SVM (linear kernel) on these features. TF-IDF + SVM achieves 68.95% accuracy and 0.75 AUC-ROC, confirming meaningful but insufficient signal from text alone.

4.3 Image-Only Baseline

ResNet-18 [4] pretrained on ImageNet is fine-tuned with `layer4` unfrozen and a new binary classification head, reusing low-level ImageNet features while adapting high-level representations to the medical domain.

4.4 Multimodal Fusion Models

Our fusion model concatenates image and text features before a learned head:

- **Image encoder**: ResNet-18 (`layer4` unfrozen), 512-dim after global average pooling.
- **Text encoder**:
  Frozen CLIP (512-dim) [11] or BioBERT `dmis-lab/biobert-base-cased-v1.2` [6] (768-dim, mean-pooled last hidden state).
- **Fusion head**: LayerNorm on each stream → concatenation → Linear(256)–ReLU–Dropout(0.5)–Linear(1).

The fused representation is:

$$z_{\text{fused}} = \left[\, \text{LN}(z_{\text{img}}) \oplus \text{LN}(z_{\text{txt}}) \,\right] \quad \text{where } z_{\text{img}} \in \mathbb{R}^{512},\ z_{\text{txt}} \in \mathbb{R}^{768}. \tag{1}$$

BioBERT is chosen over CLIP because it is pretrained on PubMed abstracts and PubMed Central texts, making it better suited to clinical terminology. We additionally fine-tune BioBERT layers 10–11 at 5×10^{-6} to adapt text representations while avoiding catastrophic forgetting.

Models are trained with `BCEWithLogitsLoss` and AdamW with separate LR groups: 10^{-4} (fusion head), 10^{-5} (ResNet `layer4`), 5×10^{-6} (BioBERT layers 10–11). Batch size 16, weight decay 10^{-4}, up to 30 epochs, early stopping (patience 7), `ReduceLROnPlateau` (factor 0.5, patience 3).

4.5 Open-Ended Captioning Pipeline

We evaluate four vision-language models on the same 769-sample ROCO-derived validation split, using BLIP variants as baselines and GPT-5 Mini as our primary system.

As baselines, we evaluate three BLIP-family models: BLIP [8] (224M parameters, also fine-tuned with frozen vision encoder on 3,084 samples), BLIP-2 [7] (OPT-2.7B backbone, Q-Former bridging frozen ViT and frozen LLM), and InstructBLIP [1] (Vicuna-7B backbone, instruction-conditioned Q-Former queries). These represent a progression in architectural complexity and serve as reference points for evaluating GPT-5 Mini's capability.

GPT-5 Mini [10] (`gpt-5-mini-2025-08-07`) is queried via the OpenAI API, with each image base64-encoded and submitted with the following system prompt:

> *"You are an expert radiologist interpreting medical images. Provide concise, accurate, and specific medical descriptions using precise clinical terminology. Focus only on what is visible in the image."*

Fine-tuning was withheld: the model weights are inaccessible for gradient updates, fine-tuning via API raises reproducibility concerns, and zero-shot performance is the relevant scenario when labeled medical data is scarce. Unlike the BLIP-family models, GPT-5 Mini generates structured multi-sentence clinical reports, a consequence of the radiologist prompt that produces the metric discrepancy discussed in Section 5.3.

Metrics. We report BLEU-4, ROUGE-1, ROUGE-L, and BERTScore [13] (P, R, F1 via DistilBERT). ROUGE measures n-gram overlap and penalises verbosity. BERTScore uses contextual embeddings

to capture semantic similarity, correctly recognising that "hepatic metastases" and "liver tumour" are clinically equivalent despite zero lexical overlap.

5 Experiments

5.1 Evaluation Metrics

Binary VQA: accuracy, macro F1, and AUC-ROC on the 959-sample validation split. Open-ended captioning: BLEU-4, ROUGE-L, and BERTScore [13].

5.2 Binary VQA Results

Table 2: Binary VQA model comparison on the validation set.

Model	Accuracy	F1	AUC-ROC
TF-IDF + Logistic Regression	0.6684	0.6441	0.7073
TF-IDF + SVM	0.6895	0.6704	0.7517
ResNet-18 (layer4 unfrozen)	0.7263	0.7500	0.7740
Fusion: ResNet-18 + CLIP	0.7579	0.7723	0.8194
Fusion: ResNet-18 + BioBERT	0.7789	0.7835	0.8204
Fusion: ResNet-18 + BioBERT (fine-tuned)	**0.7947**	**0.8000**	**0.8281**

Results show a clear progression from text-only baselines to multimodal fusion. Moving from ResNet-18 alone (72.63%) to CLIP fusion (+3.16 pp) confirms that visual features alone are insufficient without textual grounding. Replacing CLIP with BioBERT adds a further 2.1 pp, reflecting its stronger clinical vocabulary alignment. We fine-tuned the last 2 layers of BioBERT on the dataset as suggested from Prof. Lin, which improved accuracy from 77.9% to 79.5% (+1.6pp). Further improvement is limited by the dataset size (750 training samples) — the model overfits heavily (train acc 97%) regardless of regularization strategy, suggesting more labeled data would be the real bottleneck to address.

5.3 Open-Ended Captioning Results

Table 3: Open-ended captioning results on the validation set (769 samples). BLIP-2, InstructBLIP, and GPT-5 Mini are zero-shot. BERTScore uses DistilBERT embeddings. **Bold** = best per column.

Model	BLEU-4	ROUGE-1	ROUGE-L	BS-P	BS-R	BS-F1
BLIP	0.0000	0.0518	0.0427	0.6539	0.7019	0.6750
BLIP	0.0000	0.0953	0.0873	0.6812	0.7025	0.6905
BLIP-2	0.0039	0.1409	0.1251	0.7559	0.7304	0.7421
InstructBLIP	0.0058	**0.1517**	**0.1345**	0.7609	0.7353	0.7467
GPT-5 Mini	**0.0088**	0.1215	0.0943	0.7107	**0.8021**	**0.7528**

BERTScore F1 improves consistently across all five models (0.675 $\rightarrow$ 0.753), reflecting genuine gains in semantic quality at each architectural step. Fine-tuning BLIP's language decoder nearly doubles ROUGE-1 (0.052 to 0.095), confirming that domain adaptation helps even at small scale. BLIP-2 surpasses fine-tuned BLIP on all metrics despite no training on our data, showing that a larger language backbone provides more value than task-specific adaptation with limited samples.

GPT-5 Mini captures clinical meaning most completely, achieving the highest BERTScore Recall (0.802) across all models, meaning it semantically covers the reference findings more thoroughly than any fine-tuned alternative. Its lower ROUGE score (0.122 vs. InstructBLIP's 0.152) is a product of output style rather than quality: GPT-5 Mini generates structured clinical reports while ROCO references are single-sentence labels, so ROUGE penalises length asymmetry rather than inaccuracy. The lower Precision (0.711) similarly reflects additional clinical detail beyond the reference. Taken together, these results show that GPT-5 Mini is the strongest model in our evaluation when assessed on semantic fidelity, and that ROUGE alone is insufficient for evaluating models that generate richer text than their reference annotations.

5.4 Qualitative Analysis

For the binary branch, the BioBERT stream directed the classification head toward image regions relevant to the specific organ in the query, with failure cases concentrated in images with motion artifacts or fine-grained indicators exceeding the 224×224 resolution. For the open-ended branch, the following example illustrates the ROUGE–BERTScore discrepancy:

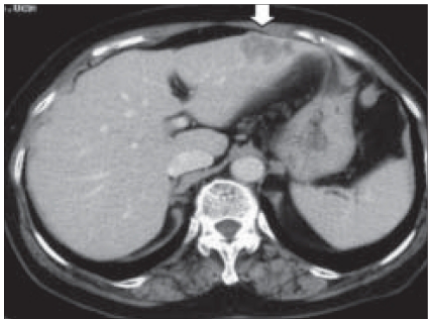

Figure 3: Sample CT image from the validation set used in the qualitative analysis.

> **Reference:** *Computed tomographic scan shows a metastatic liver tumour (arrow).*
>
> **GPT-5 Mini:** *Contrast-enhanced axial CT of the upper abdomen. Numerous well-defined hypoattenuating nodules throughout the hepatic parenchyma (several mm to >1 cm), most consistent with multifocal hepatic metastases. Peripheral/subcapsular lesion along the superior anterior hepatic surface abutting the diaphragm (arrow). Impression: multifocal liver metastases; subcapsular hepatic lesion.*

Same diagnosis, near-zero lexical overlap. ROUGE scores this near zero; BERTScore Recall of 0.802 correctly identifies high semantic coverage.

5.5 Compute Resources

Binary fusion experiments used NVIDIA T4 and A100 GPUs via Google Colab Pro, requiring $\approx$45 minutes to converge over 30 epochs (batch size 16, max sequence length 128). Open-ended experiments ran on an NVIDIA RTX 5070 Ti (17.1 GB VRAM) locally; BLIP fine-tuning completed in under 2 hours and each model took 30–45 minutes. GPT-5 Mini inference over 769 samples took $\approx$2 hours via the OpenAI API.

6 Conclusion

We presented MedVQA-Router, a two-stage pipeline for Medical VQA that routes questions to a binary classification branch or an open-ended captioning branch. Combining ResNet-18 with fine-tuned BioBERT achieves 79.5% accuracy and 0.83 AUC-ROC, outperforming all single-modality baselines by over 7 pp. The consistent advantage of BioBERT over CLIP confirms the value of domain-adapted language models for medical text understanding. For the open-ended branch, GPT-5 Mini achieves the highest BERTScore F1 (0.7528) and Recall (0.8021), while InstructBLIP leads on ROUGE, revealing that metric choice critically shapes conclusions in medical captioning research. Future work includes cross-attention fusion for finer-grained image-text interaction, end-to-end multi-task routing, and evaluation on larger benchmarks such as VQA-RAD and SLAKE [9].

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